

Requirements Table — Smart Parking Space Rental & License Validator

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall record vehicle entry time via license plate scan and compute total billing amount upon exit based on tiered hourly duration.	High	Pass: The recorded entry time and calculated bill match the actual parking duration and applicable billing tier. Fail: The entry time or billing amount is incorrect.	Required for accurate parking billing and vehicle tracking.
FR-002	Functional	The system shall allow vehicle owners to reserve an available parking bay.	High	Pass: When an available bay is selected, the system creates and displays a valid reservation. Fail: The reservation is not created or an unavailable bay is assigned.	Enables real-time parking bay reservations.
FR-003	Functional	The system shall validate a vehicle's number plate against its parking reservation at the entry gate.	High	Pass: A vehicle with a valid reservation is verified successfully. Fail: A vehicle with an invalid or missing reservation is not verified.	Prevents unauthorized vehicles from entering reserved parking spaces.
FR-004	Functional	The system shall calculate the parking fee according to the vehicle's parking duration and the applicable progressive hourly rates.	High	Pass: The calculated fee matches the applicable rate for the recorded parking duration. Fail: An incorrect rate or duration is used.	Ensures accurate and consistent parking charges.
FR-005	Functional	The system shall flag vehicles that remain in the parking area beyond their authorized parking duration.	Medium	Pass: A vehicle exceeding its authorized duration is flagged by the system. Fail: An overdue vehicle is not flagged.	Helps the parking warden identify unauthorized overstays.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional — Performance & Security	The gate camera license plate lookup API shall verify parking reservation validity in under 300 ms.	High	Pass: Benchmarking tests confirm that reservation verification completes in under 300 ms under simulated peak load. Fail: Verification takes 300 ms or longer.	Ensures fast and secure gate validation.
NFR-002	Nonfunctional — Availability	The parking system shall maintain at least 99% availability during operating hours.	High	Pass: Availability monitoring records at least 99% system uptime during the defined operating period. Fail: Recorded availability falls below 99%.	Ensures the parking service remains reliably accessible to vehicle owners and parking wardens.